



6 Statistics

Business Analytics, Jaggies et al.

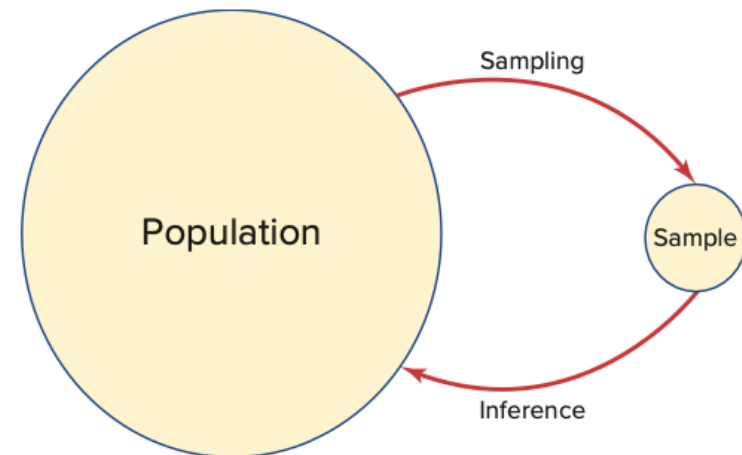
Learning Objectives (LOs)

- LO 1** Population versus statistics
- LO 2** Descriptive statistics
- LO 3** Boxplots and z-scores

Data Population and Samples

- It is not feasible to collect data that comprise a population of all elements of interest.
 - Too expensive
 - It is impossible
- A sample is a subset of the population and is used for analyses.
- Traditional statistical techniques use sample information to draw conclusions about the population.

FIGURE 1.3 Population versus sample



Introductory Case: Investment Decision (1/2)

- Dorothy Brennan works as a financial advisor at a large investment firm.
- She meets with an inexperienced investor who has some questions regarding two approaches to mutual fund investing: growth investing versus value investing.
- The investor has heard that growth funds invest in companies whose stock prices are expected to grow at a faster rate, relative to the overall stock market
- On the other hand, value funds invest in companies whose stock prices are below their true worth.
- The investor has also heard that the main component of investment return is through capital appreciation in growth funds and through dividend income in value funds.

Introductory Case: Investment Decision (2/2)

- The investor shows Dorothy the annual return data for Fidelity's Growth Index Fund (Growth) and Fidelity's Value Index fund.

Year	Growth	Value
1984	-5.50	-8.59
1985	39.91	22.10
⋮	⋮	⋮
2019	38.42	31.62

- It is difficult for the investor to draw any conclusions from the data in their present form.
- Dorothy will use the sample information for the following tasks.
 - Calculate and interpret the typical return for these two mutual funds.
 - Calculate and interpret the investment risk for these two mutual funds.

Summary Measures: *central location*

- Measures of central location attempt to find a typical or central value that describes the data.
- The **arithmetic mean** is the primary measure of central location.
 - Referred to as the mean or the average
 - Simply add up all the observations and divide by the number of observations.
- The **population mean** is denoted as μ .
 - N observations in the population: $x_1, x_2, \dots, x_N$
 - $\mu = \frac{\sum x_i}{N}$
 - μ is a **parameter**
- The **sample mean** is denoted as $\bar{x}$.
 - n observations in the sample: $x_1, x_2, \dots, x_n$
 - $\bar{x} = \frac{\sum x_i}{n}$
 - $\bar{x}$ is a **statistic**

Summary Measures

- We can also use numerical descriptive measures to extract meaningful information for data.
 - Determined values that are easy to calculate, interpret, and compare with one another.
 - Central location: a typical or central value in the data
 - We will examine the three mostly widely used measures of central location: **mean**, **median** and **mode**.
 - Then we discuss
 - Percentile: a measure of relative position.
 - Dispersion: a measure of variability.

Summary Measures: Mean, Median Mode

- **Mean** can give a misleading description of the center in the presence of extremely small or large observations, or outliers.
- **Median** can also be a measure of central location.
 - Middle value of a data set: an equal number of observations lie above and below the median
 - Arrange the data in ascending order
 - The middle value if the number of observations is odd
 - The average of the two middle values if the number of observations is even
- If the mean and median are different, it is likely the variable contains outliers.
- **Mode** of a variable is the observation that occurs most frequently.
 - There can be one or more modes

One mode: **unimodal**

Two modes: **bimodal**

Two or more: **multimodal**

Measures of Location

- Example with Excel
- Data Analysis > Descriptive Statistics.

The image shows the 'Descriptive Statistics' dialog box in Microsoft Excel. The dialog is titled 'Descriptive Statistics' and has standard window controls (minimize, maximize, close) in the top right corner. It is divided into two main sections: 'Input' and 'Output options'. In the 'Input' section, the 'Input Range' is set to '\$B\$1:\$C\$37', 'Grouped By' is set to 'Columns' (indicated by a selected radio button), and the 'Labels in first row' checkbox is checked. In the 'Output options' section, the 'Output Range' is set to 'E1', 'New Worksheet Ply' and 'New Workbook' are unselected, and the 'Summary statistics' checkbox is checked. Below this, there are three unchecked checkboxes: 'Confidence Level for Mean:' (with a value of 95%), 'Kth Largest:' (with a value of 1), and 'Kth Smallest:' (with a value of 1). On the right side of the dialog, there are three buttons: 'OK', 'Cancel', and 'Help'.

Descriptive Statistics

Input

Input Range: \$B\$1:\$C\$37

Grouped By: ☒ Columns ☐ Rows

☒ Labels in first row

Output options

☒ Output Range: E1

☐ New Worksheet Ply:

☐ New Workbook

☒ Summary statistics

☐ Confidence Level for Mean: 95 %

☐ Kth Largest: 1

☐ Kth Smallest: 1

OK Cancel Help

Measures of Location

- Example: The mean and median for the Growth and Value variables from the introductory case.

Growth		Value	
Mean	15.10742857	Mean	11.44457143
Standard Error	4.026783239	Standard Error	3.028998513
Median	14.44	Median	15.09
Mode	#N/A	Mode	#N/A
Standard Deviation	23.82277091	Standard Deviation	17.91979686
Sample Variance	567.5244138	Sample Variance	321.1191197
Kurtosis	1.064454138	Kurtosis	1.893804308
Skewness	0.02988603	Skewness	-1.007368283
Range	120.38	Range	90.6
Minimum	-40.9	Minimum	-46.52
Maximum	79.48	Maximum	44.08
Sum	528.76	Sum	400.56
Count	35	Count	35

Summary Measures: Percentile

- The median is the middle observation. Half of the observations fall below and above the median.
- The median is also called the 50th percentile.
- A percentile is technically a measure of location
 - The p th percentile divides a variable into two parts.
 - Approximately p percent of the observations are less than the p th percentile.
 - Approximately $(100-p)$ percent of the observations are greater than the p th percentile.

EXCEL: `percentile.inc()` or `percentile()`

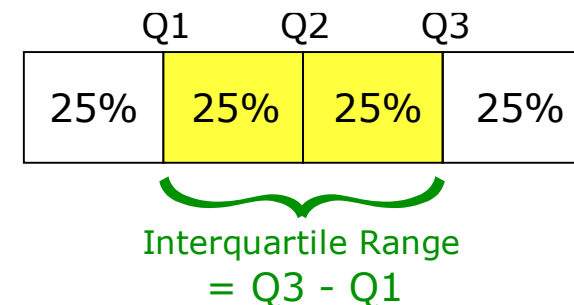
Measures of Location

- Example: The quartiles of the Growth and Value variables.

	Min	Q1	Q2	Q3	Max
Growth	-40.90	2.13	14.44	31.995	79.48
Value	-46.52	1.195	15.09	22.185	44.08

Summary Measures: Dispersion

- Measures of central location do not describe the underlying dispersion.
- Measures of dispersion gauge the variability.
 - 0 indicates all the observations are identical
 - Increases as the observations become more diverse
- The range is the simplest measure.
 - Difference between the maximum and minimum
 - Not good because it focuses solely on extreme observations
- The interquartile range (IQR) is the difference between the third quartile and the first quartile.
 - $IQR = Q_3 - Q_1$
 - The range of the middle 50% of the variable
 - Does not depend on the extreme observations



Summary Measures: Variance and Stdev

- The **variance** and the **standard deviation** are the two most widely used measures of dispersion.
 - Compute the average of the squared differences
 - The squaring of the differences emphasizes larger differences
- The **population variance** is denoted $\sigma^2 = \frac{\sum(x_i - \mu)^2}{N}$.
- The **sample variance** is denoted $s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$.
- .
- The standard deviation of each is the positive square root (σ , or s)

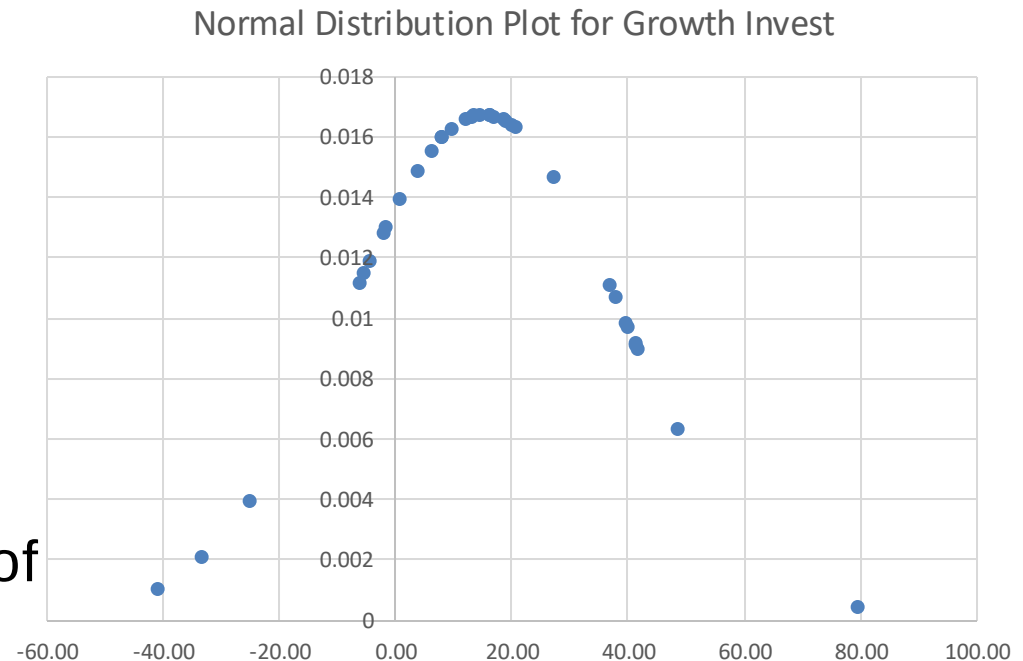
Normal distribution plot

- Based on the mean and Standard deviation of the random variable, we can generate the Normal distribution value of the data.

$$z = (x - \mu)/\sigma \text{ or } (x - \bar{x})/s$$

where $Z \sim N(0, 1)$

- The plot the PDF value of each point will signify the bell-shape curve.
- Ex: The bell-shape curve of the Growth-investment.



Measures of Dispersion and Shape

- In some instances, analysis entails comparing the variability of two or more variables that have different means or units of measurement.
- The coefficient of variation (CV) is the ratio of the standard deviation to the mean.
- The higher the coefficient of variation, the greater the level of dispersion around the mean.
 - Sample: $CV = \frac{s}{\bar{x}}$
 - Population: $CV = \frac{\mu}{\sigma}$

Measures of Dispersion and Shape

- Example: Calculate and interpret the coefficient of variation (CV) for the Growth and Value variables.
- Growth: $CV = \frac{s}{\bar{x}} = \frac{23.799}{15.755} = 1.511$
- Value: $CV = \frac{s}{\bar{x}} = \frac{17.979}{12.005} = 1.498$
- As measured by CV, the Growth variable has slightly more relative dispersion as compared to Value

Measures of Dispersion and Shape

- In general, investments with higher returns also carry higher risk.
- The average return represents an investor's reward, whereas variance, or equivalently standard deviation, corresponds to risk.
- The Sharpe ratio is the “reward-to-variability” ratio.
 - Calculated as $\frac{\bar{x}_i - R_f}{s_i}$
 - R_f is the mean return for a risk-free asset such as a Treasury bill (T-bill)
 - The numerator measures the extra reward for the added risk, and the difference is excess return
- The higher the Sharpe ratio, the better the investment compensates its investors for risk.

Measures of Dispersion and Shape

- Example: Compute the Sharpe ratios for the Growth and Value fund assuming $R_f = 2\%$.
- Because the standard deviation of Growth is greater than the standard deviation of Value, $23.799 > 17.979$, Growth is considered riskier than Value.
- Growth's Sharpe ratio: $\frac{15.755 - 2}{23.799} = 0.58$
- Value's Sharpe ratio: $\frac{12.005 - 2}{17.979} = 0.56$
- Growth provides a higher Sharpe ratio than Value ($0.58 > 0.56$); therefore, Growth offered more reward per unit of risk.

Measure of Shape: Skewness

- Measures of shape reveal whether the distribution of the variable is symmetric or if the tails are more or less extreme than the normal distribution.
- The **skewness coefficient** measures the degree to which a distribution is not symmetric about its mean.
 - Calculated as $\frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$
 - **Symmetric**: coefficient of 0
 - **Positively skewed**: positive coefficient
 - **Negatively skewed**: negative coefficient



Measure of Shape: Kurtosis

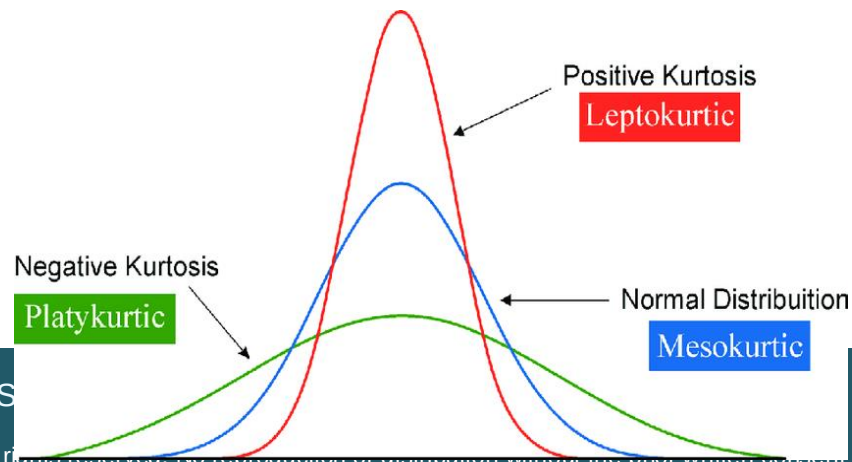
- The kurtosis coefficient is a summary measure that tells us whether the tails of the distribution are more or less extreme than the normal distribution (mesokurtic).
- A distribution that has tails that are more extreme than the normal distribution is leptokurtic (*lepto* from the Greek word for slender).
 - A return distribution is often leptokurtic, where curve is with higher peak—implying the existence of outliers.
 - If a return distribution is leptokurtic, but we assume that it is normally distributed in statistical models, then we will underestimate the likelihood of very bad or very good returns.
- A platykurtic (*platy* from the Greek word for flat) distribution is one that has lower (flatter) peak, or tails that are less extreme, than the normal distribution.

Measure of Shape: Kurtosis

- The kurtosis coefficient is calculated as:

$$\frac{n}{(n-1)(n-2)(n-3)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^4.$$

- The kurtosis coefficient of a normal distribution is 0 (adjusted from 3) (Mesokurtic).
 - Kurtosis > 0: Higher peak than a normal distribution (Leptokurtic)
 - Kurtosis < 0: Lower peak than a normal distribution (Platykurtic)



Measure of Shape: Skewness and Kurtosis

- Example: Interpret the skewness and the kurtosis coefficients for the Growth and Value variables.
- The skewness coefficient and the (excess) kurtosis coefficient for Growth are -0.029 and 0.974 , respectively.
- These values imply that the return distribution for Growth is slightly negatively skewed, and the distribution has higher peak than the normal distribution.
- With a skewness coefficient of -1.024 and a (excess) kurtosis coefficient of 1.853 , the return distribution for Value is also negatively skewed, and it has higher peak than the normal distribution, too.

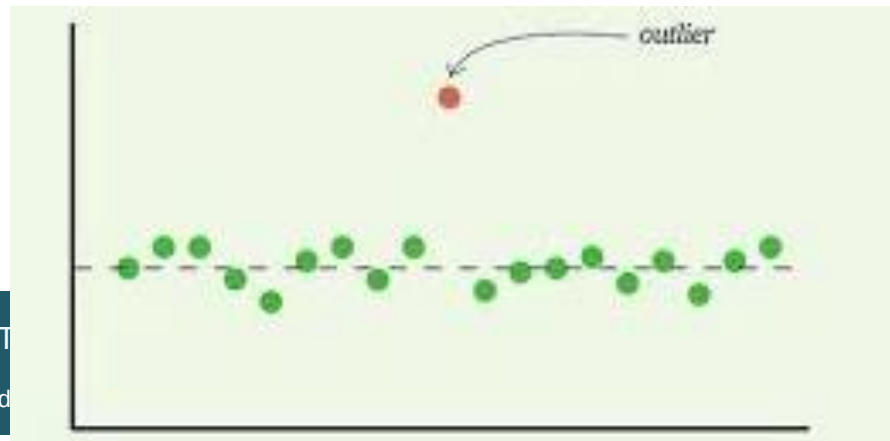
Summary Measures

- Example for the Growth and Value variables from the introductory case.

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Mean	15.10742857	Mean	11.44457143
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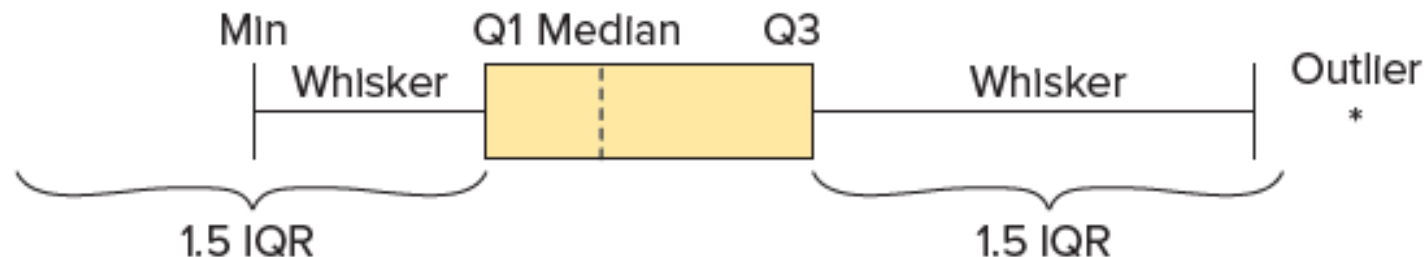
Detecting Outliers

- Extremely large or small observations for a variable are referred to as outliers
- Outliers can unduly influence summary statistics, such as the mean or the standard deviation.
 - In a small sample, the impact of outliers is particularly pronounced.
- Alternatively, outliers may indicate bad data due to incorrectly recorded observations or incorrectly included observations in the data set.
- Common methods for detecting outliers: **Boxplot** or **z-score**



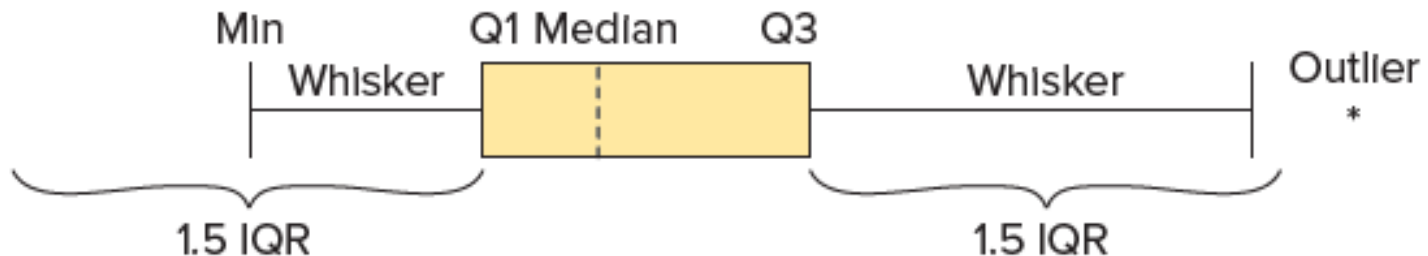
Detecting Outliers: Boxplot

- A boxplot, also referred to as a box-and-whisker plot, is a way to graphically display a five-number summary.
 - the minimum, the quartiles (Q1, Q2, and Q3), and the maximum.
 - Draw a box encompassing the first and third quartiles.
 - Draw a dashed vertical line in the box at the median.
 - Calculate the IQR. Draw a whisker that extends from Q1 to the minimum value that is not further from $1.5 \times \text{IQR}$ from Q1. Similarly, draw a line that extends from Q3 to the maximum value that is not farther than $1.5 \times \text{IQR}$ from Q3.
 - Use an asterisk (or another symbol) to indicate observations that are farther than $1.5 \times \text{IQR}$ from the box. These observations are considered outliers.



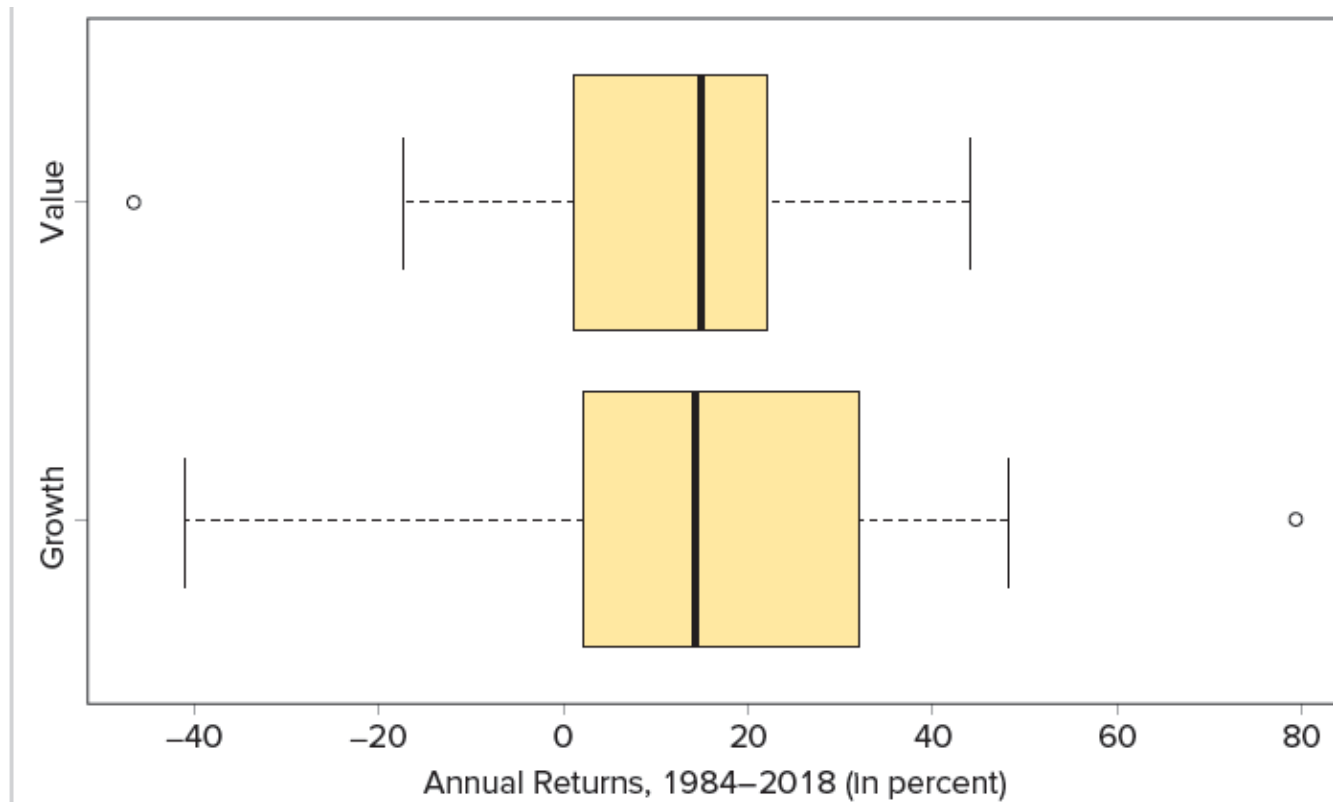
Detecting Outliers

- A boxplot is also used to informally gauge the shape of the distribution.
- Symmetry is implied if the median is in the center of the box and the left/right whiskers are equidistant from their respective quartiles.
- If the median is left of center and the right whisker is longer than the left whisker, then the distribution is positively skewed.
- Similarly, if the median is right of center and the left whisker is longer than the right whisker, then the distribution is negatively skewed.
- If outliers exist, we need to include them when comparing the lengths of the left and right whiskers.



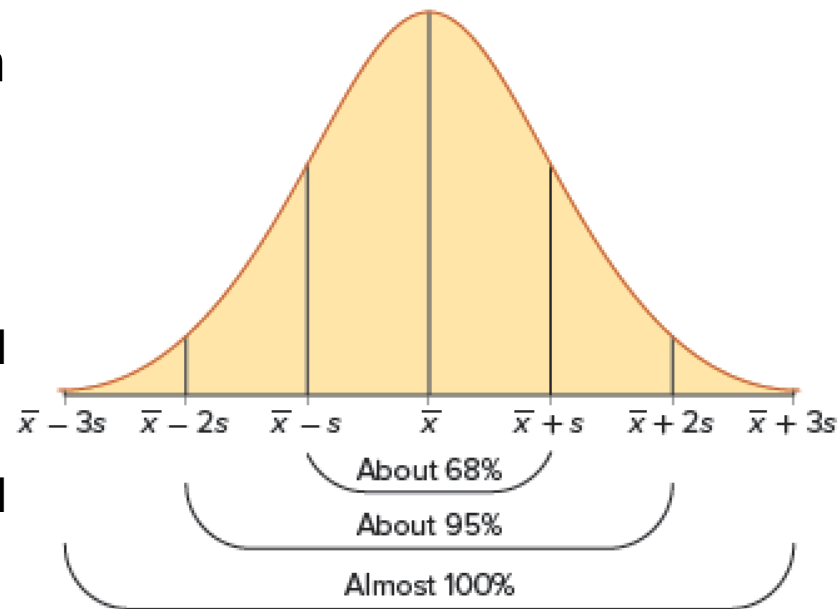
Detecting Outliers

- Example: construct a boxplot for the Growth and Value variables from the introductory case.
- Excel : Insert > Chart > Box & Whisker.



Detecting Outliers: Empirical rule

- The **empirical rule**: percentage of observations that fall within a specified number of standard deviations from the mean.
- Assume the observations are drawn from a relatively symmetric and bell-shaped distribution, perhaps by an inspection of its histogram:
 - Approximately 68% of all observations fall in the interval $\bar{x} \pm s$.
 - Approximately 95% of all observations fall in the interval $\bar{x} \pm 2s$.
 - Approximately 100% of all observations fall in the interval $\bar{x} \pm 3s$.



Detecting Outliers: z-score

- We use the z-score to find the relative position of an observation:

$$z = \frac{x - \bar{x}}{s}$$

- A z-score is a unitless measure.
- It measures the distance of an observation from the mean in terms of standard deviations.
- Converting observations into z-scores is also called standardizing the observations.
 - A common technique used in data analytics when dealing with variables measured using different scales.
- If the distribution of a variable is relatively symmetric and bell-shaped, we can also use z-scores to detect outliers.
 - Since almost all observations fall within three standard deviations of the mean, it is common to treat an observation as an outlier if its z-score is more than 3 or less than -3 .

Detecting Outliers: Compute z-score

- Example: What are the z-scores for the minimum and maximum values of the Growth and Value variables from the introductory case?

Fund	Minimum	Maximum	Mean	Standard Deviation
Growth	−40.90	79.48	15.1074	23.8228
Value	−46.52	44.08	11.4446	17.9198

- Growth minimum: $z = \frac{-40.90 - 15.1074}{23.8228} = -2.3510$
- Growth maximum: $z = 2.7021$
- Value minimum: $z = -3.2347$
- Value maximum: $z = 1.8212$
- Recall the box-and-whisker plot identified the Growth maximum and Value minimum as outliers.

Conclusion

- Data in nature is too big, samples is used instead for Data analytics
- Statistics is a tool to analyze and draw conclusion about the population:
 - descriptive stats.
- Descriptive stats can measure location, dispersion, shape and others of the sampled data.
- Boxplot is a visualization to describe descriptive stats for single or multiple samples.
- z-score is used for measuring relative position of the observation based on Normal distribution
- Outlier of data can be detected using Boxplot or z-score.